Supplementary Information for
Joint Mitigation of Algorithmic and Physical Errors in Noisy Hamiltonian Simulation

This Supplementary Information collects the technical and experimental material that supports the main text. Supplementary Note 1 introduces zero-noise extrapolation, Richardson extrapolation, the coupled one-dimensional path, the general resource theorem and the sampling estimates. Supplementary Note 2 presents the finite-order BCH truncation tool and records the resulting commutator estimates for representative Hamiltonian classes. Supplementary Notes 3–6 document the hardware circuit, readout calibration, learned sparse Pauli–Lindblad channel, hardware extrapolation data and numerical validation.

The central theoretical point is that the joint mitigation protocol does not rely on convergence of the full BCH series. Instead, the BCH expansion is used only to a finite order, and the discarded tail is controlled by doubly right-nested commutator bounds. This makes the power-series input to Richardson extrapolation rigorous in the local Lindbladian settings considered in the main text.

Supplementary Note 1 Richardson Preliminaries

1.1 Zero-noise extrapolation background

Quantum error mitigation (QEM) [1] reduces noise-induced bias in observable expectation values by post-processing data from noisy circuit executions. Unlike full quantum error correction, QEM does not require encoding the computation into a protected code space, but it can increase the number of circuit samples needed for a fixed target accuracy.

Zero-noise extrapolation (ZNE) [2, 3] is one of the standard QEM primitives. Let ρ_λ denote the output state of a circuit at effective fault rate, or noise strength, λ. For an observable O, ZNE measures expectation values at several accessible noise strengths $\lambda_1, \ldots, \lambda_M$ and fits them with an ansatz $f(\lambda; \boldsymbol{\theta})$:

$$\mathrm{Tr}[O\rho_{\lambda_m}] \approx f(\lambda_m; \boldsymbol{\theta}). \tag{S1}$$

The zero-noise value is then estimated by evaluating the fitted function at $\lambda = 0$,

$$\mathrm{Tr}[O\rho_0] \approx f(0; \boldsymbol{\theta}^*). \tag{S2}$$

In the perturbative regime, the dependence on λ can be modeled by a low-degree polynomial,

$$f(\lambda; \boldsymbol{\theta}) = \sum_{\ell=0}^{M-1} \theta_\ell \frac{\lambda^\ell}{\ell!}. \tag{S3}$$

When the number of sampled noise strengths equals the number of polynomial coefficients, Eq. (S3) is a Richardson extrapolation in the noise direction. More general overdetermined fits are also common in experiments.

The noise strength can be amplified in several ways. Unitary folding replaces a gate or circuit U by $U(U^\dagger U)^n$, increasing the effective fault rate by a factor of approximately $1+2n$ without requiring a detailed noise model [4–6]. In near-term Hamiltonian simulation circuits, however, folding also increases depth. An alternative, used in the hardware workflow here, is to learn an effective sparse Pauli–Lindblad channel and sample additional Pauli insertions to realize calibrated noise strengths without lengthening the target Trotter circuit [7].

1.2 Richardson nodes and coefficients

Richardson extrapolation is a standard way to remove the leading terms of a polynomial error expansion. The following node choice and coefficient bound are used both for ordinary Trotter-step extrapolation and for the coupled one-dimensional path introduced below.

Lemma S1 (Richardson nodes and coefficients, see [8, Lemma A.9])**.** *Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ can be decomposed as*

$$f(x) = P_m(x) + R_m(x), \tag{S4}$$

where P_m is a degree-$(m-1)$ polynomial and R_m is the remainder. Given $t > 0$ and $s \in (0,1)$, choose

$$K = \max\left\{\frac{m}{\pi}, \frac{2t}{s}\right\}, \qquad r_j = \left\lceil \frac{K}{\sin^2(\pi(2j-1)/8m)} \right\rceil, \tag{S5}$$

and

$$s_j = \frac{t}{r_j}, \qquad b_j = \prod_{\ell \neq j} \frac{1}{1 - r_\ell / r_j}. \tag{S6}$$

Then $1 \leq \|\boldsymbol{b}\|_1 \leq C \log m$ for an absolute constant C, $r_j = O(\max\{m^3, m^2 t/s\}/j^2)$, and $\max_j s_j \leq s$. The m-term Richardson extrapolation

$$F^{(m)}(s) = \sum_{j=1}^{m} b_j f(s_j) \tag{S7}$$

satisfies

$$\left| F^{(m)}(s) - f(0) \right| \leq \|\boldsymbol{b}\|_1 \max_j |R_m(s_j)|. \tag{S8}$$

For a two-dimensional tensor-product extrapolation with step-size coefficients b_j and independent noise-direction coefficients d_ℓ, the coefficient one-norm becomes $\|\boldsymbol{b}\|_1 \|\boldsymbol{d}\|_1$. Thus each grid point must be estimated more accurately by the extra factor $\|\boldsymbol{d}\|_1$ compared with a one-dimensional path using the same step-size nodes. For generic noise-direction nodes x_ℓ, the zero-noise Richardson coefficients obey the Lagrange expression

$$\|\boldsymbol{d}\|_1 = \sum_\ell \left| \prod_{q \neq \ell} \frac{x_q}{x_q - x_\ell} \right|. \tag{S9}$$

If the noise-direction nodes are poorly conditioned, the statistical prefactor can dominate the apparent simplicity of a rectangular extrapolation grid.

1.3 The coupled one-dimensional path

The main text studies expectation values

$$g(s,\lambda) = \mathrm{Tr}\Big[O\left(\widetilde{\mathcal{P}}(sT,\lambda)^{1/s}\right)\rho_0\Big], \tag{S10}$$

where s is the product-formula step size, T is the total simulated time, λ is the effective physical-noise strength and O is the measured observable. The desired noiseless, zero-step value is

$$g(0,0) = \mathrm{Tr}\Big[Oe^{-i\,\mathrm{ad}_H T}\rho_0\Big]. \tag{S11}$$

One could extrapolate in two independent variables: first send $\lambda \to 0$ at fixed s, and then send $s \to 0$. The joint protocol instead restricts the data to a one-dimensional path

$$f(s) = g(s,\lambda(s)), \qquad \lambda(s) = c(sT)^{p+1}, \tag{S12}$$

for a p-th order product formula. Because $\lambda(0) = 0$, the path has the correct limit $f(0) = g(0,0)$.

The exponent in Eq. (S12) matches the first nonzero deterministic product-formula correction. A p-th order product formula has leading Trotter bias of order s^{p+1}, while the leading Markovian noise contribution is linear in λ. If $\lambda(s)$ decayed with a smaller exponent, physical noise would dominate the leading bias. If it decayed with a larger exponent, the Trotter contribution would remain the bottleneck and the noise would be suppressed more strongly than is needed for the extrapolation. The matched choice places both error mechanisms in the same polynomial expansion in s, so the same Richardson coefficients cancel them simultaneously.

1.4 General one-dimensional resource theorem

This subsection records the general resource estimate underlying the one-dimensional joint extrapolation protocol. The class-dependent work is to bound the finite-BCH tail parameter $\bar{\alpha}_{\mathrm{comm},q_0}(s)$ and the effective-generator coefficient scale μ_{comm,q_0}. These quantities are defined from the truncated BCH generator in Supplementary Note 2; the theorem below only uses their role in controlling the Richardson remainder.

Theorem S2 (General joint extrapolation resource bound)**.** *Let $H = \sum_{\gamma=1}^{\Gamma} H_\gamma$ be simulated by a p-th order product formula*

$$P(t) = \prod_{v=1}^{M} e^{-ita_v H_{\gamma_v}}. \tag{S13}$$

Assume that the implemented layer is followed by noise $e^{\lambda\mathcal{L}_v}$ and that the effective noise strength is tunable above an intrinsic floor $\lambda_{\min}$. Fix $c > 0$ and set $\lambda(s) = c(sT)^{p+1}$.

Let q_0 be a BCH truncation order. Let $\bar{\alpha}_{\mathrm{comm},q_0}(s)$ be the finite-BCH truncation parameter of Theorem S6, evaluated for the generators

$$\mathcal{K}_{2v-1} = -isTa_v\,\mathrm{ad}_{H_{\gamma_v}}, \qquad \mathcal{K}_{2v} = c(sT)^{p+1}\mathcal{L}_v, \tag{S14}$$

and let μ_{comm,q_0} be a bound on the coefficients of the truncated effective generator after the substitution $\lambda(s) = c(sT)^{p+1}$.

There is an algorithm that estimates

$$\mathrm{Tr}[Oe^{-i\,\mathrm{ad}_H T}\rho_0] \tag{S15}$$

to additive error $\varepsilon\|O\|$ using

$$O\left(\max\{\log^3(1/\varepsilon), \log^2(1/\varepsilon)s_0^{-1}\}\frac{(\log\log(1/\varepsilon))^3}{\varepsilon^2}\right) = \widetilde{O}\left(\frac{1}{s_0\varepsilon^2}\right) \tag{S16}$$

total Trotter steps. The number of independent circuit runs is $\widetilde{O}(1/\varepsilon^2)$, and each run contains at most

$$O\big(\max\{\log^3(1/\varepsilon), \log^2(1/\varepsilon)s_0^{-1}\}\big) = \widetilde{O}(s_0^{-1}) \tag{S17}$$

Trotter steps.

Here $m = O(\log(1/\varepsilon))$ is the Richardson order, chosen so that $2e^{1-m} \le \varepsilon/(4C\log m)$, and s_0 is the largest step size satisfying

$$\begin{aligned} s_0(2\mu_{\mathrm{comm},q_0}T)^{1+1/p} &\le e^{-1}, \\ 2e\bar{\alpha}_{\mathrm{comm},q_0}(s_0) &\le s_0\frac{\varepsilon}{4eC\log m}. \end{aligned} \tag{S18}$$

The statement assumes that the intrinsic floor is compatible with the smallest Richardson node,

$$\lambda_{\min} = \widetilde{O}(c(s_0T)^{p+1}). \tag{S19}$$

Proof. Define the one-variable extrapolation function

$$f(s) = \mathrm{Tr}\left[O\left(\widetilde{\mathcal{P}}(sT, c(sT)^{p+1})\right)^{1/s}\rho_0\right]. \tag{S20}$$

By the finite-order BCH construction in Supplementary Note 2, the exact noisy evolution can be compared with the truncated effective evolution $e^{\mathcal{G}_{(q_0)}(s)T}$. The latter has an expansion

$$e^{\mathcal{G}_{(q_0)}(s)T} = e^{-i\,\mathrm{ad}_H T} + \sum_{q=p}^{\infty}\widetilde{\mathcal{E}}_{(q_0),q}s^q, \tag{S21}$$

with coefficients bounded by

$$\|\widetilde{\mathcal{E}}_{(q_0),q}\| \le e(2\mu_{\mathrm{comm},q_0}T)^{q(1+1/p)} \tag{S22}$$

in the regime $2\mu_{\mathrm{comm},q_0}T \ge 1$; the case $2\mu_{\mathrm{comm},q_0}T \le 1$ is easier.

Write $f(s) = P_m(s) + R_m(s)$, where P_m keeps the terms in Eq. (S21) through degree $m-1$ and R_m includes both the higher-order series tail and the finite-BCH truncation error. If s obeys Eq. (S18), then the geometric tail bound and Theorem S6 give

$$|R_m(s)| \le \frac{\varepsilon}{2C\log m}\|O\|. \tag{S23}$$

The function $\bar{\alpha}_{\mathrm{comm},q_0}(s)/s$ is monotone for $s \ge 0$ because it is a polynomial with non-negative coefficients and lowest degree at least one after division by s. Hence Eq. (S23) holds for all Richardson nodes $s_j \le s_0$.

Applying Lemma S1 gives

$$\left|\sum_{j=1}^{m} b_j f(s_j) - f(0)\right| \le \|\boldsymbol{b}\|_1 \max_j |R_m(s_j)| \le \frac{\varepsilon}{2}\|O\|. \tag{S24}$$

The remaining error budget is statistical. Estimating each $f(s_j)$ to accuracy $\varepsilon\|O\|/(2\|\boldsymbol{b}\|_1)$ with failure probability $O(1/m)$ requires

$$O(\|\boldsymbol{b}\|_1^2\varepsilon^{-2}\log m) \tag{S25}$$

independent executions at that node. Since $1/s_j = r_j = O(\max\{m^3, m^2 s_0^{-1}\}/j^2)$, the maximum coherent Trotter depth is Eq. (S17) and summing r_j over nodes gives Eq. (S16). Finally, the smallest node satisfies $\min_j c(s_jT)^{p+1} = \widetilde{\Omega}(c(s_0T)^{p+1})$, which gives Eq. (S19). ∎

1.5 Variance-sensitive sampling refinement

The preceding theorem uses a bounded-outcome concentration estimate. A sharper statement keeps the actual single-shot variances of the extrapolation data points. This refinement does not change the worst-case $1/\varepsilon^2$ dependence, but it makes explicit the statistical prefactor saved when a one-dimensional path replaces a two-dimensional grid.

Lemma S3 (Bernstein bound with shot allocation)**.** *Let*

$$\widehat{F} = \sum_{i=1}^{m} a_i \widehat{O}_i, \qquad \widehat{O}_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} X_{i,\ell}, \tag{S26}$$

where the random variables $X_{i,\ell}$ are independent, $\mathbb{E}X_{i,\ell} = O_i$, $\mathrm{Var}(X_{i,\ell}) = \sigma_i^2$ and $|X_{i,\ell}| \le \|O\|$. Then

$$\Pr\{|\widehat{F} - \mathbb{E}\widehat{F}| \ge \varepsilon\} \le 2\exp\left(-\frac{\varepsilon^2}{2V + \frac{4}{3}\|O\|\varepsilon B}\right), \tag{S27}$$

where

$$V = \sum_{i=1}^{m} \frac{a_i^2 \sigma_i^2}{n_i}, \qquad B = \max_i \frac{|a_i|}{n_i}. \tag{S28}$$

With equal shots at each point, a sufficient total number of shots is

$$N_{\mathrm{eq}} = O\left(\left(\frac{m \sum_i a_i^2 \sigma_i^2}{\varepsilon^2} + \frac{\|O\| m \max_i |a_i|}{\varepsilon}\right) \log \frac{1}{\delta}\right). \tag{S29}$$

With variance-sensitive shot allocation, one obtains

$$N_{\mathrm{opt}} = O\left(\left(\frac{(\sum_i |a_i| \sigma_i)^2}{\varepsilon^2} + \frac{\|O\| \|\boldsymbol{a}\|_1}{\varepsilon}\right) \log \frac{1}{\delta}\right). \tag{S30}$$

Proof. Apply Bernstein's inequality to the centered variables $a_i(X_{i,\ell} - O_i)/n_i$. Their total variance is V, and each summand has magnitude at most $2\|O\||a_i|/n_i$. This gives Eq. (S27). For equal shots, $n_i = N_{\mathrm{tot}}/m$, so $V = m \sum_i a_i^2 \sigma_i^2 / N_{\mathrm{tot}}$ and $B = m \max_i |a_i| / N_{\mathrm{tot}}$, which yields Eq. (S29). For optimized allocation, split the budget into a variance part with $n_i^{(v)} \propto |a_i| \sigma_i$ and a range-control part with $n_i^{(r)} \propto |a_i|$. Cauchy–Schwarz gives $V \le (\sum_i |a_i| \sigma_i)^2 / N_v$, while the range part gives $B \le \|\boldsymbol{a}\|_1 / N_r$. Choosing N_v and N_r to make the two Bernstein denominator terms $O(\varepsilon^2 / \log(1/\delta))$ proves Eq. (S30). □

For the one-dimensional joint path, set $a_j = b_j$ and let $\sigma_{j,\mathrm{path}}^2 = \mathrm{Var}(X_{s_j, \lambda(s_j)})$. Then

$$N_{\mathrm{1D,eq}} = O\left(\left(\frac{m \sum_j b_j^2 \sigma_{j,\mathrm{path}}^2}{\varepsilon^2} + \frac{\|O\| m \max_j |b_j|}{\varepsilon}\right) \log \frac{1}{\delta}\right), \tag{S31}$$

and

$$N_{\mathrm{1D,opt}} = O\left(\left(\frac{(\sum_j |b_j| \sigma_{j,\mathrm{path}})^2}{\varepsilon^2} + \frac{\|O\| \|\boldsymbol{b}\|_1}{\varepsilon}\right) \log \frac{1}{\delta}\right). \tag{S32}$$

For a two-dimensional tensor-product extrapolation with step-size coefficients b_j and noise-direction coefficients d_ℓ, the estimator coefficients are $a_{j,\ell} = b_j d_\ell$. With $\sigma_{j,\ell}^2 = \mathrm{Var}(X_{s_j, \lambda_\ell})$,

$$N_{\mathrm{2D,eq}} = O\left(\left(\frac{m_s m_\lambda \sum_{j,\ell} b_j^2 d_\ell^2 \sigma_{j,\ell}^2}{\varepsilon^2} + \frac{\|O\| m_s m_\lambda \max_{j,\ell} |b_j d_\ell|}{\varepsilon}\right) \log \frac{1}{\delta}\right), \tag{S33}$$

and

$$N_{\text{2D,opt}} = O\left(\left(\frac{(\sum_{j,\ell}|b_j d_\ell|\sigma_{j,\ell})^2}{\varepsilon^2} + \frac{\|O\|\|\boldsymbol{b}\|_1\|\boldsymbol{d}\|_1}{\varepsilon}\right)\log\frac{1}{\delta}\right). \tag{S34}$$

Thus the two-dimensional grid pays both the additional number of circuit settings and the coefficient weights in the noise direction. In the random-site Pauli estimator for average magnetization, one shot has $X \in \{\pm 1\}$ and variance $1-\mu^2$, where $\mu = \mathbb{E}X$. A sufficient condition for the one-dimensional path to have no larger equal-shot variance proxy than the two-dimensional grid is

$$\sum_j b_j^2(1-\mu_{j,\text{path}}^2) \leq \sum_{j,\ell} b_j^2 d_\ell^2 (1-\mu_{j,\ell}^2), \tag{S35}$$

with the analogous optimized-allocation condition obtained by replacing squared coefficients with absolute coefficients and variances with standard deviations.

1.6 Technical lemmas

Let $\mathcal{A}(\tau)$ and $\mathcal{B}(\tau)$ be two continuous operator-valued functions. We use the following interaction-picture identity to separate a time-ordered exponential generated by $\mathcal{A}(\tau)+\mathcal{B}(\tau)$ into an evolution under $\mathcal{A}(\tau)$ and a conjugated perturbation under $\mathcal{B}(\tau)$.

Lemma S4 ([9, Page 21])**.** *Let $\mathcal{H}(\tau) = \mathcal{A}(\tau) + \mathcal{B}(\tau)$ be an operator-valued function defined for $\tau \in \mathbb{R}$ with continuous summands $\mathcal{A}(\tau)$ and $\mathcal{B}(\tau)$. Then*

$$\begin{aligned}
&\exp_{\mathcal{T}}\left(\int_0^t \mathcal{H}(\tau)\,\mathrm{d}\tau\right) \\
&\quad = \exp_{\mathcal{T}}\left(\int_0^t \mathcal{A}(\tau)\,\mathrm{d}\tau\right)\exp_{\mathcal{T}}\left(\int_0^t \exp_{\mathcal{T}}^{-1}\left(\int_0^{\tau_1}\mathcal{A}(\tau_2)\,\mathrm{d}\tau_2\right)\mathcal{B}(\tau_1)\exp_{\mathcal{T}}\left(\int_0^{\tau_1}\mathcal{A}(\tau_2)\,\mathrm{d}\tau_2\right)\,\mathrm{d}\tau_1\right).
\end{aligned}$$

The next elementary stability estimate is used when replacing powers of an exact channel by powers of an approximate effective channel.

Lemma S5 ([10, Lemma 3])**.** *Let $\mathcal{N}_1$ and $\mathcal{N}_2$ be two superoperators, where $\mathcal{N}_1$ is a quantum channel. For any positive integer k,*

$$\left\|\mathcal{N}_1^k - \mathcal{N}_2^k\right\| \leq k\max\{1, \|\mathcal{N}_2\|\}^{k-1}\|\mathcal{N}_1 - \mathcal{N}_2\|. \tag{S36}$$

Supplementary Note 2 Finite-Order BCH Truncation as the Analytic Tool

Let $\mathcal{K}_1, \dots, \mathcal{K}_M$ be general Lindbladians or superoperators. The BCH formula can be written as

$$e^{\mathcal{K}_M}\cdots e^{\mathcal{K}_2}e^{\mathcal{K}_1} = \exp\left(\sum_{q=1}^{\infty}\Phi_q\right), \tag{S37}$$

where each Φ_q is a weighted sum of right-nested commutators of $\mathcal{K}_j$ [11]:

$$\Phi_q = \sum_{\substack{p_v \geq 0 \\ p_1+\cdots+p_M=q}} \frac{1}{p_1!\cdots p_M!}\sum_{\sigma\in S_q} c_{\sigma,q}\mathcal{R}_\sigma\{[\underbrace{\mathcal{K}_M,\dots,\mathcal{K}_M}_{p_M},\dots,\underbrace{\mathcal{K}_1,\dots,\mathcal{K}_1}_{p_1}]\}. \tag{S38}$$

Here S_q is the set of permutations of $[q]$, $\mathcal{R}_\sigma$ permutes the q entries of the right-nested commutator, and

$$c_{\sigma,q} = \frac{1}{q^2}\frac{(-1)^{d_\sigma}}{\binom{q-1}{d_\sigma}}, \qquad |c_{\sigma,q}| \le \frac{1}{q^2}, \tag{S39}$$

where d_σ is the number of descents of σ.

Define the sum of q-fold right-nested commutator norms

$$\alpha_{\text{comm}}^{(q)} = \sum_{v_1,\ldots,v_q=1}^{M} \left\| [\mathcal{K}_{v_1},\ldots,\mathcal{K}_{v_q}] \right\|. \tag{S40}$$

Using Eq. (S38), one has $\|\Phi_q\| \le q^{-2}\alpha_{\text{comm}}^{(q)}$ [12]. Hence the full BCH series is guaranteed to converge if there exist constants $J, C \ge 0$ such that

$$\sup_{q\ge J} \alpha_{\text{comm}}^{(q)} \le C. \tag{S41}$$

A direct bound gives

$$\alpha_{\text{comm}}^{(q)} \le \left(2\sum_{v=1}^{M}\|\mathcal{K}_v\|\right)^q, \tag{S42}$$

by $\|[A,B]\| \le 2\|A\|\|B\|$ and the triangle inequality.

For local product formulas, Eq. (S42) misses the relevant structure. If each $\mathcal{K}_v$ is composed of k-local terms and the total interaction strength touching any site is g, the commutator structure yields the sharper fixed-order scaling

$$\alpha_{\text{comm}}^{(q)} = O(Nq!(2kg)^q), \tag{S43}$$

rather than $(2Ng)^q$. This fixed-order estimate is the source of commutator-sensitive Trotter bounds for local Hamiltonians.

The difficulty is that advanced Trotter extrapolation analyses require more than a bound at one fixed order: they require a controlled expansion of the extrapolation function near $s = 0$. For a k-local system, the product-formula generators scale as $\mathcal{K}_v = O(s)$, so the commutator estimate behaves as

$$\alpha_{\text{comm}}^{(q)} = O(s^q q!(kg)^q). \tag{S44}$$

The factorial growth dominates the s^q suppression at large q, so the full BCH series need not have a positive radius of convergence for general k-local systems. Consequently, the power expansion used in an extrapolation proof would be only a formal asymptotic series if it were based on convergence of the infinite BCH series. This is the gap addressed by the finite-order truncation theorem: we keep the BCH generator only up to a chosen order q_0 and bound the discarded tail directly.

Following Ref. [10], the relevant tail parameter is expressed using doubly right-nested commutator sums. For positive integers $q_1,\ldots,q_d$, define

$$\bar{\alpha}_{\text{comm}}^{(q_1,\ldots,q_d)} = \sum_{v_{1,1},\ldots,v_{d,q_d}=1}^{M} \left\| \Big[[\mathcal{K}_{v_{1,1}},\ldots,\mathcal{K}_{v_{1,q_1}}],\ldots,[\mathcal{K}_{v_{d,1}},\ldots,\mathcal{K}_{v_{d,q_d}}] \Big] \right\|. \tag{S45}$$

These quantities are tailored to the finite BCH remainder. For local systems, the nested support can grow only when a new local term overlaps the current support, leading to products of local factors and support-growth factors rather than a global norm bound.

Theorem S6 ([10, Theorem 6], BCH truncation bound)**.** *For any positive integer q_0, define*

$$\bar{\alpha}_{\mathrm{comm},q_0} = \bar{\alpha}_{\mathrm{comm}}^{(q_0+1)} + \sum_{r=0}^{\infty} \frac{1}{(r+1)!} \sum_{\substack{1\le p_1,\ldots,p_{r+1}\le q_0 \\ p_1+\cdots+p_{r+1}\ge q_0+2}} \bar{\alpha}_{\mathrm{comm}}^{(p_1,\ldots,p_{r+1})}. \tag{S46}$$

If $\bar{\alpha}_{\mathrm{comm},q_0} \le 1$, then the truncation error of the q_0-th order BCH formula obeys

$$\left\| \exp\left(\sum_{q=1}^{q_0} \Phi_q \right) - \prod_{v=1}^{M} e^{\mathcal{K}_v} \right\| \le e\bar{\alpha}_{\mathrm{comm},q_0}. \tag{S47}$$

This theorem is used in the joint extrapolation analysis at the level of one noisy product-formula step. If the noiseless product formula has factors indexed by v, we write the Hamiltonian part and the coupled noise part as

$$\mathcal{K}_{2v-1} = -isTa_v \, \mathrm{ad}_{H_{\gamma_v}}, \qquad \mathcal{K}_{2v} = c(sT)^{p+1}\mathcal{L}_v. \tag{S48}$$

The single-step channel is a product of exponentials of these generators. Instead of treating Eq. (S37) as a convergent infinite identity, we define the truncated scaled generator

$$\mathcal{G}_{(q_0)}(s) = \frac{1}{sT} \sum_{q=1}^{q_0} \Phi_q\big(sT, c(sT)^{p+1}\big)\,. \tag{S49}$$

The Richardson remainder is then controlled by two ingredients: the finite polynomial expansion of $\mathcal{G}_{(q_0)}(s)$ and the truncation error in Theorem S6. The order q_0 is chosen so that the BCH tail is below the target accuracy while keeping the commutator parameters in the effective-generator expansion under control.

We next record how the general truncation theorem is instantiated for the Hamiltonian classes used in the main analysis.

2.1 k-local Lindbladian systems

Suppose that each generator $\mathcal{K}_v$ is a k-local superoperator on a size-N lattice $\Lambda = [N]$:

$$\mathcal{K}_v = \sum_{\gamma=1}^{\Gamma} \mathcal{K}_\gamma^{(v)}, \qquad |\operatorname{supp}(\mathcal{K}_\gamma^{(v)})| \le k. \tag{S50}$$

We say that $\mathcal{K} = \sum_v \mathcal{K}_v$ is g-extensive if, for every site j,

$$\sum_{v=1}^{M} \sum_{\gamma:\, j\in \operatorname{supp}(\mathcal{K}_\gamma^{(v)})} \|\mathcal{K}_\gamma^{(v)}\| \le g. \tag{S51}$$

Sparse Pauli–Lindblad generators naturally fit this framework. If

$$\mathcal{L} = \mathcal{H} + \sum_{v=1}^{M} \mathcal{D}[D_v], \tag{S52}$$

where $H = \sum_\gamma h_\gamma P_\gamma$ contains Pauli strings of weight at most k and each jump operator $D_v = \sum_\eta d_{v,\eta} Q_{v,\eta}$ contains Pauli strings of weight at most k, then the Hamiltonian part is k-local as a superoperator and the dissipative part is at most $2k$-local. With

$$g_H = \max_j \sum_{\gamma:\, j \in \mathrm{supp}(P_\gamma)} |h_\gamma|, \tag{S53}$$

and

$$g_D = \max_j \sum_v \sum_{\eta,\lambda:\, j \in \mathrm{supp}(Q_{v,\eta}) \cup \mathrm{supp}(Q_{v,\lambda})} |d_{v,\eta} d^*_{v,\lambda}|, \tag{S54}$$

one obtains an extensiveness parameter $G = 2g_H + 2g_D$. Thus the abstract k-local superoperator result applies to sparse Pauli–Lindbladians after replacing k by $2k$ and g by G.

Lemma S7 (Right-nested local commutator bound)**.** *Let $\mathcal{K}_1, \ldots, \mathcal{K}_q$ be superoperators on a size-N lattice such that each $\mathcal{K}_j$ is k-local and g_j-extensive. Then*

$$\sum_{v_1,\ldots,v_q} \big\| [\mathcal{K}_{1,v_1}, \ldots, \mathcal{K}_{q,v_q}] \big\| \le N(q-1)!(2k)^{q-1} \prod_{j=1}^{q} g_j. \tag{S55}$$

Theorem S8 (Doubly nested bound for k-local superoperators)**.** *Let $\mathcal{K} = \sum_v \mathcal{K}_v$ be k-local and g-extensive, and let $\mathcal{A}$ have support size at most k. For positive integers $q_1, \ldots, q_d$, define*

$$P_r = 1 + \sum_{j=r}^{d} q_j, \qquad P_{d+1} = 1. \tag{S56}$$

Then the doubly nested commutator with one inserted local superoperator satisfies

$$\begin{aligned} &\sum_{v_{1,1},\ldots,v_{d,q_d}} \Big\| \Big[[\mathcal{K}_{v_{1,1}}, \ldots, \mathcal{K}_{v_{1,q_1}}], \ldots, [\mathcal{K}_{v_{d,1}}, \ldots, \mathcal{A}, \ldots, \mathcal{K}_{v_{d,q_d}}] \Big] \Big\| \\ &\qquad \le \left(\prod_{r=1}^{d} P_{r+1} q_r! (2kg)^{q_r} \right) \|\mathcal{A}\|. \end{aligned} \tag{S57}$$

Proof sketch. Decompose every $\mathcal{K}_v$ into its k-local constituents. The innermost commutator is bounded using Lemma S7. Moving outward, a new nested layer contributes only when at least one of its local terms overlaps the current support. If the current support size is at most kP_r, the g-extensiveness condition bounds the overlapping norm sum by $kP_r g$. Iterating this argument from the innermost layer to the outermost layer gives the product in Eq. (S57). □

Lemma S9 (Finite BCH approximation for k-local systems)**.** *Given k-local and g-extensive superoperators $\mathcal{K}_v$, for any $\varepsilon \in (0, 1/2)$ and $q_0 \ge \log(2N/\varepsilon)$, if*

$$4e^2 k g q_0 \le 1, \tag{S58}$$

then

$$\left\| \exp\left(\sum_{q=1}^{q_0} \Phi_q \right) - \prod_{v=1}^{M} e^{\mathcal{K}_v} \right\| \le \varepsilon. \tag{S59}$$

Lemma S9 is the operational replacement for full BCH convergence. The convergence condition based on Eq. (S42) would force $\sum_v \|\mathcal{K}_v\| = O(1)$, which corresponds to $g = O(1/N)$ in local systems. By contrast, the finite truncation bound remains useful in the larger regime $g = O(1/(k \log N))$, where the infinite BCH series is not guaranteed to converge.

Lemma S10 (Effective-generator coefficient scale for k-local systems)**.** *Let* $\mathrm{ad}_H = \sum_{\gamma\in[\Gamma]} \mathcal{A}_\gamma$ *and* $\mathcal{L}_{\mathrm{tot}} = \sum_{v=1}^{M} \mathcal{L}_v$ *be* k*-local, with* ad_H g_H*-extensive and* $\mathcal{L}_{\mathrm{tot}}$ g_L*-extensive. Let the* p*-th order product formula use* $M = \Upsilon\Gamma$ *layers and coefficients bounded by* $a_{\max}$*. Then the coefficient scale in Theorem S2 can be chosen as*

$$\mu_{\mathrm{comm},q_0} \leq q_0 N^{1/(p+1)} \left(2a_{\max} k \Upsilon g_H + (2ckg_L)^{1/(p+1)}\right). \tag{S60}$$

Proof sketch. Group each right-nested commutator according to the number q_1 of Hamiltonian generators and the number q_2 of noise generators. The k-local commutator bound gives a factor $N(q_1+q_2-1)!(2k)^{q_1+q_2-1}(\Upsilon g_H)^{q_1} g_L^{q_2}$, and there are $\binom{q_1+q_2}{q_1}$ choices for the Hamiltonian positions. After substituting $\lambda(s) = c(sT)^{p+1}$, the grade is $q_1 + (p+1)q_2$. Taking the corresponding root and using $q_1 + q_2 \leq q_0$ gives Eq. (S60). □

Theorem S11 (Joint extrapolation for k-local systems)**.** *Assume the setting of Lemma S10. Suppose that the intrinsic noise strength satisfies*

$$\lambda_{\min} = \widetilde{O}\left(\frac{1}{(Nk\Upsilon a_{\max} g_H T)^{1+1/p} k g_L}\right). \tag{S61}$$

Then there is an estimator for $\mathrm{Tr}[Oe^{-i\,\mathrm{ad}_H T}\rho_0]$ *with additive error* $\varepsilon\|O\|$ *using*

$$O\left(\max\{\log^3(1/\varepsilon), \Xi \log^2(1/\varepsilon)\} \frac{(\log\log(1/\varepsilon))^3}{\varepsilon^2}\right) \tag{S62}$$

total Trotter steps, where

$$\Xi = N^{1/p}(k\Upsilon a_{\max} g_H T)^{1+1/p} \log^{1+1/p}(N/\varepsilon). \tag{S63}$$

The number of independent circuit runs is $\widetilde{O}(1/\varepsilon^2)$*, and each run contains at most*

$$O\big(\max\{\log^3(1/\varepsilon), \Xi \log^2(1/\varepsilon)\}\big) \tag{S64}$$

Trotter steps.

Proof. Choose

$$q_0 = \left\lceil \log \frac{8e^2 N \log m}{\varepsilon} \right\rceil = O(\log(N/\varepsilon)). \tag{S65}$$

For the coupled generators, the total local extensiveness entering the finite-BCH bound is

$$\Upsilon sT a_{\max} g_H + c(sT)^{p+1} g_L. \tag{S66}$$

Applying Lemma S9 gives

$$\bar{\alpha}_{\mathrm{comm},q_0}(s) \leq 2N\left(4eq_0 k(\Upsilon sTa_{\max} g_H + c(sT)^{p+1} g_L)\right)^{q_0+1} \tag{S67}$$

whenever the factor in parentheses is at most $1/2$.

Set

$$c = \frac{(k\Upsilon a_{\max} g_H)^{p+1}}{k g_L}. \tag{S68}$$

Then $(kcg_L)^{1/(p+1)} = k\Upsilon a_{\max} g_H$, so the Hamiltonian and noise contributions are balanced in Eq. (S60). The step size selected by Eq. (S18) satisfies

$$s_0^{-1} = O\Big(N^{1/p}(k\Upsilon a_{\max} g_H T)^{1+1/p} \log^{1+1/p}(N/\varepsilon)\Big) = O(\Xi). \tag{S69}$$

Substituting this value of s_0 into Theorem S2 yields Eq. (S62). The smallest coupled noise strength is $\widetilde{\Omega}(c(s_0T)^{p+1})$, which gives the feasibility condition in Eq. (S61). □

In the noiseless limit $g_L = 0$, Theorem S11 reduces to a commutator-sensitive resource estimate for ordinary Trotter-step extrapolation. With local noise satisfying Eq. (S61), the choice of c in Eq. (S68) rescales s_0 only by a constant factor compared with the noiseless analysis. Thus physical-noise mitigation is incorporated into the Trotter extrapolation with constant asymptotic overhead, while retaining the optimal worst-case $\widetilde{O}(1/\varepsilon^2)$ sampling dependence required by general QEM lower bounds [13, 14].

2.2 Geometrically local observables

For geometrically local Hamiltonians, the same theorem can be localized around the support of the measured observable. Let $\Lambda \subseteq \mathbb{Z}^d$ be a d-dimensional lattice, and suppose that

$$\mathrm{ad}_H = \sum_{x\in\Lambda} \mathcal{A}_x, \tag{S70}$$

where each $\mathcal{A}_x$ is supported inside an ℓ_1 ball $B_R(x)$. Let O be supported on $B_{R_O}(x_O)$ and define

$$J_H = \max_{x\in\Lambda} \sum_{\gamma\in\mathcal{I}_x} \|\mathcal{A}_\gamma\|. \tag{S71}$$

If $|B_R(x)| \le C_d R^d$, then ad_H is $g_H = O(C_d R^d J_H)$-extensive.

For any $l \ge 0$, define the truncated Hamiltonian by retaining only terms centered in the ball

$$B_{R_O+l+R}(x_O). \tag{S72}$$

A Lieb–Robinson-type stability estimate [15] gives

$$\left\| e^{i\,\mathrm{ad}_H\, t} O - e^{i\,\mathrm{ad}_{H_{(l)}}\, t} O \right\| \le |\operatorname{supp}(O)| \|O\| \min\{e^{-l/R}(e^{eC_dR^dt} - 1), 1\}. \tag{S73}$$

Choosing

$$l = eC_dR^{d+1}T + R\log(2|\operatorname{supp}(O)|/\varepsilon) \tag{S74}$$

makes the truncation error at time T at most $\varepsilon\|O\|/2$.

The joint extrapolation theorem is then applied to the truncated Hamiltonian $H_{(l)}$ and the restricted noise generator on the enlarged region $B_{R_O+l+2R}(x_O)$. The effective system size is

$$N_{\text{eff}} = |B_{R_O+l+2R}(x_O)| = O\Big(C_d(R_O + eC_dR^{d+1}T + R\log(|\operatorname{supp}(O)|/\varepsilon) + R)^d\Big). \tag{S75}$$

Replacing N by N_{eff} in Theorem S11 gives an estimator for the truncated dynamics with error at most $\varepsilon\|O\|/2$. Together with Eq. (S73), this yields an $\varepsilon\|O\|$-accurate estimator for the original local observable. The dependence on the total lattice size is therefore replaced by the volume of the effective light cone around $\operatorname{supp}(O)$.

2.3 Second-quantized electronic structure

The same truncation theorem also applies to second-quantized electronic-structure Hamiltonians in the plane-wave dual basis. Following Ref. [16], write

$$H = \underbrace{\sum_{j,k=1}^{n} t_{j,k} A_j^\dagger A_k}_{T} + \underbrace{\sum_{j=1}^{n} u_j N_j}_{U} + \underbrace{\sum_{j,k=1}^{n} g_{j,k} N_j N_k}_{V} . \tag{S76}$$

Here $A_j^\dagger, A_j$ are fermionic creation and annihilation operators and $N_j = A_j^\dagger A_j$. The coefficient vectors obey

$$\|\vec{t}\|_\infty = O(1/n), \quad \|\vec{t}\|_1 = O(n), \quad \|\vec{u}\|_\infty = O(n), \quad \|\vec{u}\|_1 = O(n^2), \tag{S77}$$

and

$$\|\vec{g}\|_\infty = O(1), \qquad \|\vec{g}\|_1 = O(n^2). \tag{S78}$$

Let $H = \sum_{v=1}^{M} H_v$ denote the decomposition into the summands of Eq. (S76). The right-nested commutators satisfy

$$\sum_{v_1,\ldots,v_q=1}^{M} \|[H_{v_1}, \ldots, H_{v_q}]\| = O(q! n^q), \tag{S79}$$

for $q \geq 2$, and each such commutator has at most q layers in the sense of Ref. [16]. Setting $\mathcal{K}_v = -i\,\mathrm{ad}_{H_v}$ and using $[\mathrm{ad}_X, \mathrm{ad}_Y] = \mathrm{ad}_{[X,Y]}$ gives the corresponding Liouvillian estimate.

Lemma S12 (Doubly nested bound for plane-wave electronic structure)**.** *Let $\mathcal{K} = \sum_{v=1}^{M} \mathcal{K}_v$ with $\mathcal{K}_v = -i\,\mathrm{ad}_{H_v}$ for the decomposition in Eq. (S76). For positive integers $q_1, \ldots, q_d$, define $P_r = \sum_{j=r}^{d} q_j$ and $P_{d+1} = 1$. Then*

$$\bar{\alpha}_{\mathrm{comm}}^{(q_1,\ldots,q_d)} = O\left(\prod_{r=1}^{d} P_{r+1} q_r! n^{q_r} \right) . \tag{S80}$$

The proof uses the same outward induction as in the local case, but the support-growth factor is replaced by the layer-counting recursion for the second-quantized operators. This provides the commutator-tail input needed by Theorem S6 and extends the finite-BCH approach beyond geometrically local lattice Hamiltonians.

Supplementary Note 3 Circuit Mapping and Readout Calibration

The hardware experiment uses a connected 9-qubit subgraph of a 13-qubit superconducting processor. The simulated observable is the average longitudinal magnetization,

$$O = \frac{1}{n} \sum_{i=1}^{n} Z_i. \tag{S81}$$

Each second-order symmetric Trotter step consists of transverse-field rotations and an Ising interaction layer. The interaction graph is decomposed into disjoint R_{ZZ} patterns so that gates within one pattern are executed in parallel. The main text shows the circuit and topology; this note records the experimental workflow and the calibration data used before the learned-noise step.

Supplementary Fig. S1 illustrates the learning-and-mitigation workflow. Target Trotter circuits are mapped to the hardware topology and compiled into native gates. Cliffordized training circuits preserve the same compiled layout and layer grouping, while replacing rotation angles by Clifford angles so that noiseless references can be computed classically. Hardware measurements of these training circuits, together with the noiseless Clifford references, determine a layerwise sparse Pauli–Lindblad model used for noise amplification and ZNE.

Before learning the effective gate-noise channel, we mitigate readout error at the observable level. Supplementary Fig. S2 shows both the full confusion matrix on the 9-qubit subgraph and the local single-qubit calibration matrices. The global matrix is strongly diagonally dominant, indicating that the readout channel is close to an independent bit-flip model on the relevant device subgraph. The local matrices capture the dominant contribution for the magnetization observable while avoiding the sampling cost of inverting the full global matrix in every data-processing step.

Supplementary Note 4 Learned Sparse Pauli–Lindblad Channel

The learned-channel quantum error mitigation workflow assumes an effective sparse Pauli–Lindblad model for each relevant circuit layer [7],

$$\mathcal{L}(\rho) = \sum_{k\in\mathcal{K}} \lambda_k (P_k \rho P_k - \rho), \tag{S82}$$

where $\mathcal{K}$ contains the Pauli components retained by the sparse model. The corresponding channel can be written as a product of Pauli dephasing components,

$$\Lambda(\rho) = \prod_{k\in\mathcal{K}} \left(w_k \rho + (1-w_k) P_k \rho P_k\right), \qquad w_k = \frac{1+e^{-2\lambda_k}}{2}. \tag{S83}$$

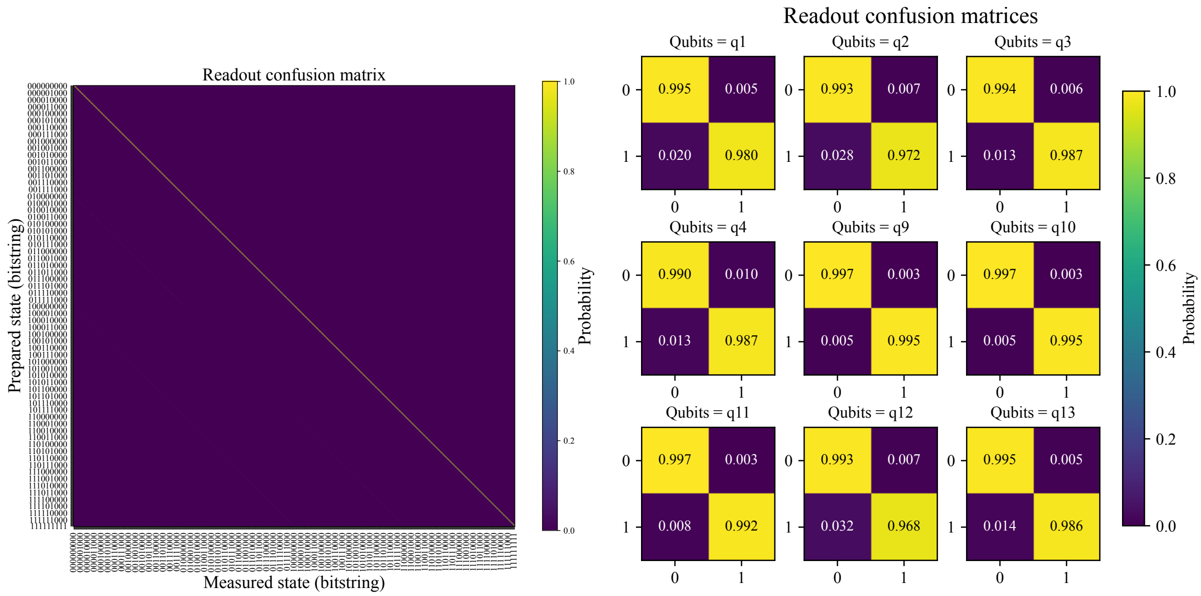
Training circuits are produced by replacing the target circuit rotation angles by Clifford angles in $\{0, \pi/2, \pi, 3\pi/2\}$ while preserving the same circuit layout, qubit mapping, layer grouping and native-gate compilation. Their noiseless expectation values are computed classically, while their noisy expectation values are measured on hardware after readout mitigation. Solving the resulting sparse inverse problem yields the effective Pauli–Lindblad rates. The learned channel is an effective model for the observable and circuit family: components that are invisible to the measured expectation or degenerate with other components need not be separately identified. This is why the main text treats the model as a calibrated noise knob rather than as microscopic noise tomography.

Supplementary Fig. S3 shows a representative learned hardware noise channel for a Trotterized circuit with step number $N_{\text{Trotter}} = 3$. The figure displays the inferred Pauli-noise components across the Clifford layers of the compiled circuit, giving a layer-resolved view of the channel used to generate controlled noise strengths for ZNE.

Supplementary Note 5 Hardware ZNE and Trotter Extrapolation

After learning the effective noise channel, we use random Pauli insertions to implement the calibrated noise strengths. For each fixed Trotter step number, the amplified-noise expectation values are fitted as functions of the noise strength and extrapolated to the zero-noise limit. In the hardware data we compare low-order polynomial fits with exponential fitting families; the latter are natural for Pauli–Lindblad channels because the channel eigenvalues depend exponentially on the learned rates. Supplementary Fig. S4 shows hardware ZNE data for two representative evolution settings. In both cases, the measured expectation values vary smoothly with the noise strength,

Figure S1: Experimental learning-and-mitigation workflow. Target Trotter circuits are mapped to the hardware topology and compiled into native gates. Cliffordized training circuits preserve the same layer structure while replacing rotation angles by Clifford angles. Hardware measurements of these training circuits, together with noiseless Clifford references, determine an effective layerwise sparse Pauli–Lindblad model used for noise amplification and ZNE. The workflow keeps the training circuits structurally matched to the target Trotter circuits so that the learned channel is tied to the compiled hardware implementation.



(a)

(b)

Figure S2: Readout calibration data for the selected 9-qubit device subgraph. (a) Full $2^9 \times 2^9$ global confusion matrix, whose entries give the conditional probabilities of measured bitstrings given prepared computational-basis states. (b) Single-qubit local calibration matrices used for observable-level readout mitigation of the magnetization. The diagonal dominance of the calibration data supports local observable-level correction for the measured average magnetization.

and the zero-noise estimates are consistent with the corresponding noiseless simulation references within the accuracy of the fit.

The hardware ZNE stage removes the physical-noise bias at fixed Trotter step number. It does not remove the deterministic error from the finite-step product formula. For this reason, the ZNE-corrected estimates are subsequently extrapolated in $x = 1/N_{\text{Trotter}}$. For the second-order symmetric formula, the leading product-formula corrections have an even-power structure in x. Supplementary Fig. S5 shows the corresponding even-polynomial Trotter extrapolation and the residual error after extrapolation. The zero-step estimates are close to the independently computed non-Trotter reference.

Supplementary Note 6 Numerical Learned-Channel Validation

To check that the learning procedure scales beyond the 9-qubit hardware instance, we also use classically simulated noisy circuits. The simulations use the same Trotterized Ising structure and average magnetization observable as in the hardware experiments. A synthetic sparse Pauli noise model is applied to the circuit, and LCQEM is run using finite-shot Clifford training data. The reference tensor-network simulations in the main text use `quimb` [17].

Supplementary Fig. S6 shows a representative learned-noise validation on a 5×6 lattice. The inferred noise parameters reproduce the noisy target-circuit expectation values under several noise strengths. This comparison isolates the learning step: the "realistic" noisy expectations are generated directly from the synthetic noise model, whereas the learned expectations are generated from the fitted sparse Pauli–Lindblad channel.

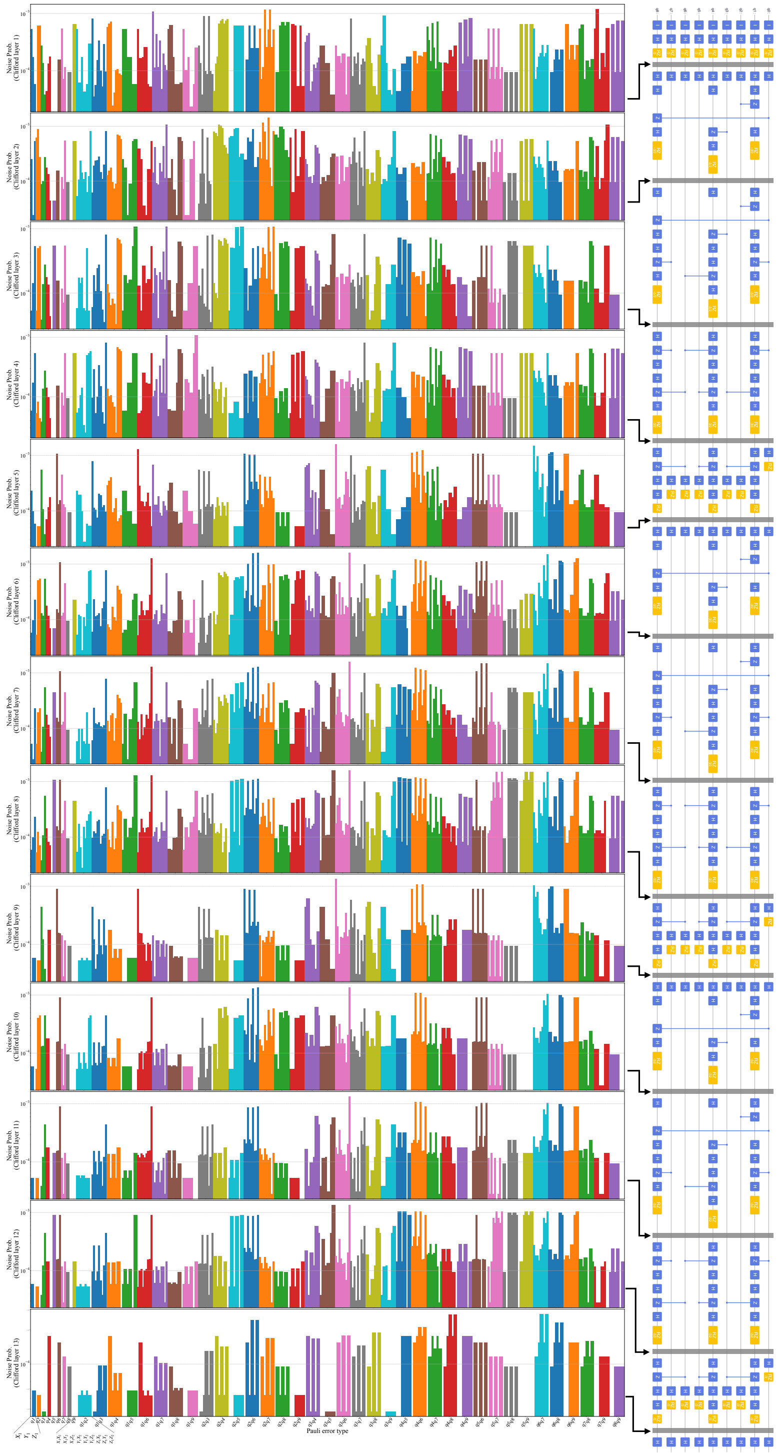
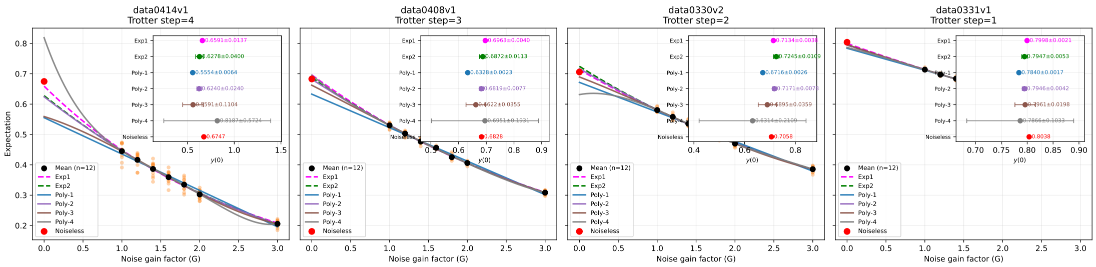
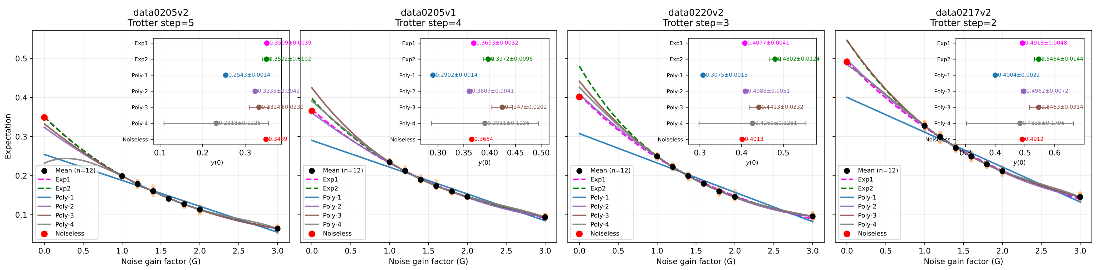


Figure S3: Learned effective hardware noise model for a Trotterized Ising circuit with Trotter step number $N_{\mathrm{Trotter}} = 3$. The plot shows the inferred sparse Pauli–Lindblad components associated with the Clifford layers in the compiled hardware circuit. These learned components provide the calibrated noise channel used for controlled noise amplification in the subsequent ZNE step.

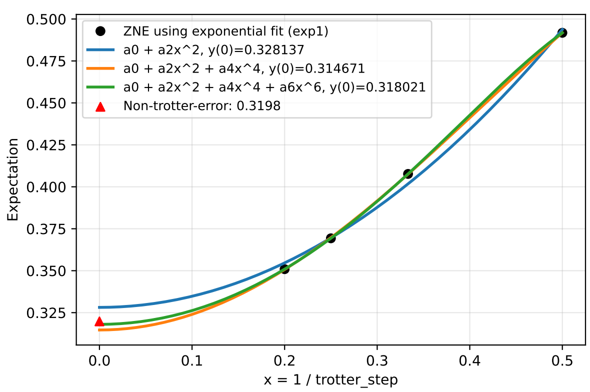


(a)

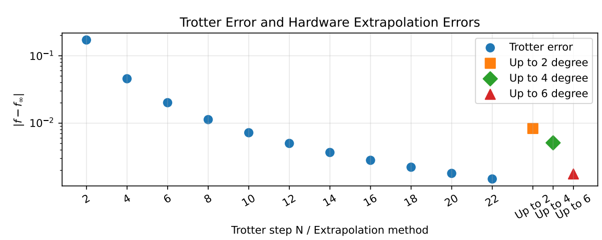


(b)

Figure S4: Hardware zero-noise extrapolation using the learned Pauli–Lindblad channel. (a) Evolution setting $r_x T = r_{zz} T = -1$. (b) Evolution setting $r_x T = r_{zz} T = -2$. The extrapolated value at zero noise is obtained by fitting the measured dependence on the noise strength. These data provide the finite-Trotter, zero-noise estimates that are subsequently used for Trotter-step extrapolation.

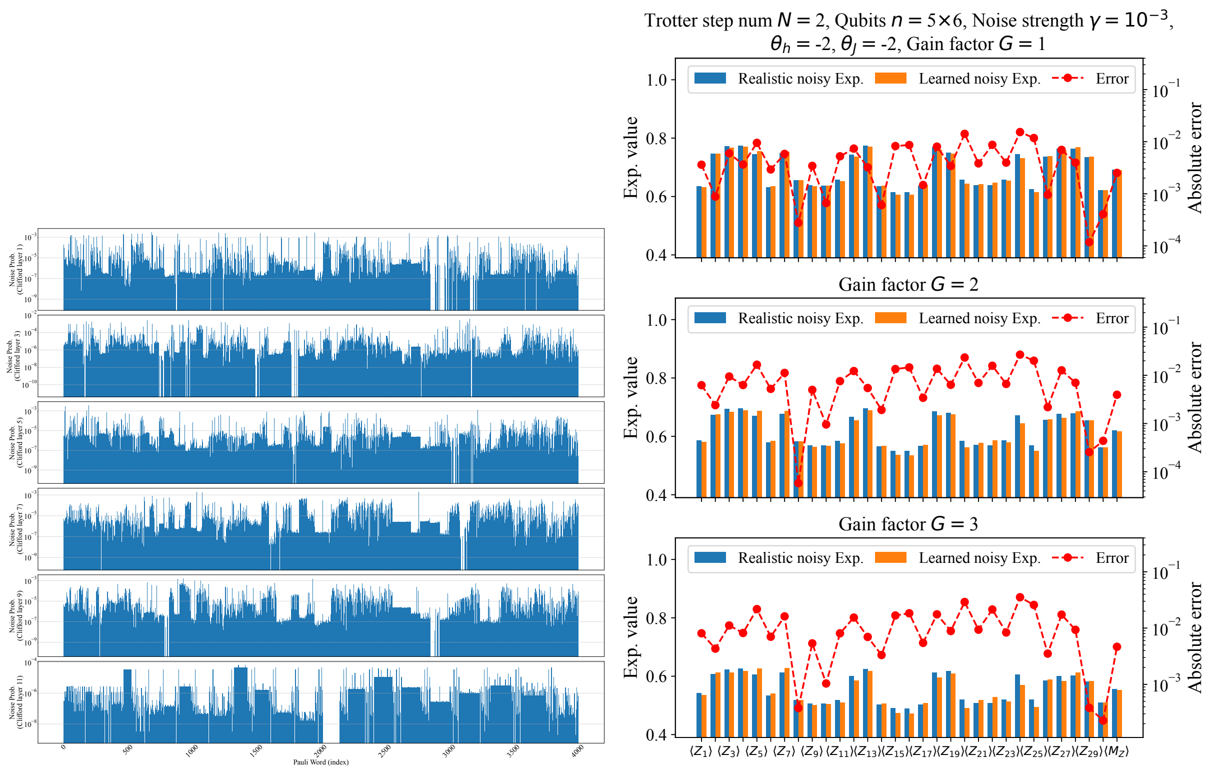


(a)



(b)

Figure S5: Trotter-step extrapolation after noise mitigation for the setting $r_x T = r_{zz} T = -2$. (a) Extrapolation of the corrected expectation value as a function of $x = 1/N_{\text{Trotter}}$ using even-polynomial fits. (b) Comparison between finite-step Trotter errors and the residual errors after zero-step extrapolation.



(a) (b)

Figure S6: Numerical validation of the learned sparse Pauli–Lindblad channel. (a) Learned noise-channel parameters for a representative 5×6 Trotter circuit instance. (b) Comparison between noisy expectation values obtained directly from the synthetic noise model and those generated from the learned model. Agreement across noise strengths indicates that the learned channel captures the observable-relevant noise components.

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